

PRACTICE REPORT

COMPUTER NETWORKS

Lab 1

COMPUTER NETWORKS

Teacher: **Phan Trung Phát**

Group: **CS4283.Q12.CTTT**

Name	ID
Nguyễn Dương Phúc	24521386

OTHER EVALUATIONS (*):

Content	Result
Average total time to complete the lab ⁽¹⁾	8 hours
Video Link ⁽²⁾ (if any)	
Feedback ⁽³⁾ (if any) + Challenges + Suggestions ...	
Self-Assessment Score ⁽⁴⁾	9/10

(*): Sections (1) and (4) are mandatory.

TASK ASSIGNMENT TABLE:

Members	Task Details
Nguyễn Dương Phúc	Question 1, 2, 3, 4, 5, 6, 7, 8, 9, expand (Work - 100%)

The section below is the detailed report of the group/individual responsible.



FOR INTERNAL CIRCULATION ONLY



Question 1

1. Evidence:

Exercise 1:

Wireshark packet capture for Exercise 1. The packet list shows a packet at time 11.029798 with details: Ethernet II, Internet Protocol Version 6, User Datagram Protocol, and Data (124 bytes). The packet details pane shows the frame structure. The packet capture file is 24521386-Exercise1.pcapng, and the total number of packets is 4251.

Exercise 2:

Wireshark packet capture for Exercise 2. The packet list shows a packet at time 7.684559 with details: Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Domain Name System (query). The packet details pane shows the frame structure. The packet capture file is 24521386-Exercise2.pcapng, and the total number of packets is 1387.

2. Explanation:

Exercise 1:

Total packet capture time: **11.029798 (s)**

Total number of packets captured: **4251 (packets)**

Exercise 2:

Total packet capture time: **7.684559 (s)**

Total number of packets captured: **1387 (packets)**

Question 2

1. Evidence:



Exercise 1:

http				
No.	Time	Source	Destination	
1	3027 7.398894	192.168.191.52	128.119.245.12	
2	3290 7.707396	128.119.245.12	192.168.191.52	
3	3383 7.814183	192.168.191.52	128.119.245.12	
4	3470 8.141764	128.119.245.12	192.168.191.52	

Packets: 4251 Displayed: 4 (0.1%)

Exercise 2:

No.	Time	Source	Destination	Protocol
1	409 3.892189	192.168.191.52	199.231.164.68	HTTP
2	452 4.161206	199.231.164.68	192.168.191.52	HTTP
3	454 4.179327	192.168.191.52	199.231.164.68	HTTP
4	457 4.180002	192.168.191.52	199.231.164.68	HTTP
5	458 4.180052	192.168.191.52	199.231.164.68	HTTP
6	470 4.460294	199.231.164.68	192.168.191.52	HTTP
7	472 4.460381	199.231.164.68	192.168.191.52	HTTP
8	478 4.461524	192.168.191.52	199.231.164.68	HTTP
9	479 4.461629	192.168.191.52	199.231.164.68	HTTP
10	481 4.479630	199.231.164.68	192.168.191.52	HTTP
11	485 4.484044	192.168.191.52	199.231.164.68	HTTP
12	529 4.743157	199.231.164.68	192.168.191.52	HTTP
13	539 4.750089	199.231.164.68	192.168.191.52	HTTP
14	541 4.770825	199.231.164.68	192.168.191.52	HTTP
15	1123 6.230715	192.168.191.52	199.231.164.68	HTTP
16	1169 6.533899	199.231.164.68	192.168.191.52	HTTP

Packets: 1387 Displayed: 16 (1.2%)

2. Explanation:

Exercise 1:

There are **4 packets** belonging to the HTTP protocol (No. **3027, 3290, 3383, and 3470**).

Exercise 2:

There are **16 packets** belonging to the HTTP protocol. (No. **409, 452, 454, 457, 458, 470, 472, 478, 479, 481, 485, 529, 539, 541, 1123, and 1169**)

Question 3

1. Evidence:

Exercise 1:



Protocol
<u>UDP</u>
<u>TCP</u>
<u>QUIC</u>
<u>TLSv1.3</u>
<u>HTTP</u>

Exercise 2:

Protocol
<u>HTTP</u>
<u>DNS</u>
DNS
DNS
<u>TCP</u>
DNS
DNS
DNS
<u>TCP</u>
DNS
DNS
DNS
<u>QUIC</u>
<u>QUIC</u>
<u>TCP</u>
<u>TCP</u>
<u>TCP</u>
<u>TLSv1.3</u>

2. Explanation:

Exercise 1: 5 protocols

- UDP
- TCP
- QUIC
- TLS
- HTTP

Exercise 2: 5 protocols

- HTTP
- DNS
- TCP
- QUIC
- TLS

Main function of those protocols

UDP: A connectionless transport protocol used for fast data transmission where reliability is less critical, such as DNS or video streaming.

TCP: A connection-oriented and reliable transport protocol that ensures accurate data delivery between client and server, commonly used in HTTP and HTTPS.

QUIC: A modern transport protocol developed by Google, built on top of UDP and integrating TLS encryption to provide faster and more reliable connections (used in HTTP/3).

TLS: A cryptographic protocol that provides secure communication over a computer network, used to encrypt data in HTTPS connections.

HTTP: An application-layer protocol used for transferring web content such as HTML pages, images, and other resources between the browser and the web server.

DNS: An application-layer protocol that translates domain names into IP addresses so devices can locate each other on the Internet. It is often used over UDP (port 53) for fast queries, and sometimes over TCP for tasks like zone transfers.

Question 4

1. Evidence:

Exercise 1:

No.	Time	Source	Destination	Protocol	Length	Info
3027	7.398894	192.168.191.52	128.119.245.12	HTTP	550	GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
3290	7.707396	128.119.245.12	192.168.191.52	HTTP	492	HTTP/1.1 200 OK (text/html)
3383	7.814183	192.168.191.52	128.119.245.12	HTTP	496	GET /favicon.ico HTTP/1.1
3470	8.141764	128.119.245.12	192.168.191.52	HTTP	538	HTTP/1.1 404 Not Found (text/html)

Exercise 2:

No.	Time	Source	Destination	Protocol	Length	Info
409	3.892189	192.168.191.52	199.231.164.68	HTTP	528	GET /faqs/ HTTP/1.1
452	4.161206	199.231.164.68	192.168.191.52	HTTP	372	HTTP/1.1 200 OK (text/html)
454	4.179327	192.168.191.52	199.231.164.68	HTTP	426	GET /style/faqs.css HTTP/1.1

2. Explanation:

Exercise 1:

The first HTTP GET packet is **No. 3027** (appears before others and requests the main HTML page from the server)

Exercise 2:

The first HTTP GET packet is **No. 409**

The basic purpose of an HTTP GET packet: to retrieve data from a server using a specified URI (Uniform Resource Identifier). It requests a resource without modifying any data on the server. For example, when a user accesses a webpage, the browser sends a GET request to download the page content.

Question 5

1. Evidence:

Exercise 1:

3027	7.398894	192.168.191.52	128.119.245.12	HTTP	550 GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
3290	7.707396	128.119.245.12	192.168.191.52	HTTP	492 HTTP/1.1 200 OK (text/html)
3383	7.814183	192.168.191.52	128.119.245.12	HTTP	496 GET /favicon.ico HTTP/1.1
3470	8.141764	128.119.245.12	192.168.191.52	HTTP	538 HTTP/1.1 404 Not Found (text/html)

\r\n
Packet No.3027 (first GET)
 [Response in frame: 3290]
 [Full request URI: <http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html>]

\r\n
Packet No.3290 (response packet)
 [Request in frame: 3027]
 [Time since request: 0.308502000 seconds]
 [Request URI: /wireshark-labs/INTRO-wireshark-file1.html]
 [Full request URI: <http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html>]

Exercise 2:

409	3.892189	192.168.191.52	199.231.164.68	HTTP	528 GET /faqs/ HTTP/1.1
452	4.161206	199.231.164.68	192.168.191.52	HTTP	372 HTTP/1.1 200 OK (text/html)
454	4.179327	192.168.191.52	199.231.164.68	HTTP	426 GET /style/faqs.css HTTP/1.1

Packet No.409 (first GET)
 [Response in frame: 452]
 [Full request URI: <http://www.faqs.org/faqs/>]

Packet No.452 (Response packet)
 [Request in frame: 409]

2. Explanation:

Exercise 1:

The response packet corresponding to the first HTTP GET is **No.3290** (In the *Hypertext Transfer Protocol* section of packet 3290, it shows “**Request in frame 3027**”, and in packet 3027, it shows “**Response in frame 3290**”).

Exercise 2:

The response packet corresponding to the first HTTP GET is **No.452** (In the *Hypertext Transfer Protocol* section of packet 452, it shows “**Request in frame 409**”, and in packet 409, it shows “**Response in frame 452**”).

Question 6

1. Evidence:

Exercise 1:

No.	Time	Source	Destination	Protocol	Length	Info
3027	7.398894	192.168.191.52	128.119.245.12	HTTP	550	GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
3290	7.707396	128.119.245.12	192.168.191.52	HTTP	492	HTTP/1.1 200 OK (text/html)
3383	7.814183	192.168.191.52	128.119.245.12	HTTP	496	GET /favicon.ico HTTP/1.1
3470	8.141764	128.119.245.12	192.168.191.52	HTTP	538	HTTP/1.1 404 Not Found (text/html)

[Time since request: 0.308502000 seconds]

Exercise 2:



No.	Time	Source
409	3.892189	192.168.191.52
452	4.161206	199.231.164.68
454	4.170327	192.168.191.52

[Time since request: 0.269017000 seconds]

2. Explanation:

Exercise 1:

Time of GET packet (No. 3027): **7.398894 (s)**

Time of 200 OK packet (No. 3290): **7.707396 (s)**

⇒ The total response time is $7.707396 - 7.398894 = \mathbf{0.308502 (s)}$

Exercise 2:

Time of GET packet (No. 409): **3.892189 (s)**

Time of 200 OK packet (No. 452): **4.161206 (s)**

⇒ The total response time is $4.161206 - 3.892189 = \mathbf{0.269017 (s)}$

Question 7

1. Evidence:

Exercise 1:

```
Line-based text data: text/html (3 lines)
<html>\n
Congratulations! You've downloaded the first Wireshark lab file!\n
</html>\n
```

01a0	3e 0a 43 6f 6e 67 72 61 74 75 6c 61 74 69 6f 6e	>.Congra tulation
01b0	73 21 20 20 59 6f 75 27 76 65 20 64 6f 77 6e 6c	s! You' ve downl
01c0	6f 61 64 65 64 20 74 68 65 20 66 69 72 73 74 20	oaded th e first
01d0	57 69 72 65 73 68 61 72 6b 20 6c 61 62 20 66 69	Wireshar k lab fi
01e0	6c 65 21 0a 3c 2f 68 74 6d 6c 3e 0a	le!-</ht ml>.

2. Explanation:

Exercise 1:

Yes, it can be found under the “**Line-based text data**” section of the **HTTP 200 OK response packet (No.3290)**.

It can also be seen in the **Packet Contents window**, where it is displayed in both **ASCII** and **hexadecimal** formats.

Question 8

1. Evidence:



Exercise 1:

No.	Time	Source	Destination	Protocol	Length	Info
3027	7.398894	192.168.191.52	128.119.245.12	HTTP	550	GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
3290	7.707396	128.119.245.12	192.168.191.52	HTTP	492	HTTP/1.1 200 OK (text/html)
3383	7.814183	192.168.191.52	128.119.245.12	HTTP	496	GET /favicon.ico HTTP/1.1
3470	8.141764	128.119.245.12	192.168.191.52	HTTP	538	HTTP/1.1 404 Not Found (text/html)

Exercise 2:

No.	Time	Source	Destination	Protocol	Length	Info
409	3.892189	192.168.191.52	199.231.164.68	HTTP	528	GET /faqs/ HTTP/1.1
452	4.161206	199.231.164.68	192.168.191.52	HTTP	372	HTTP/1.1 200 OK (text/html)
454	4.170327	192.168.191.52	199.231.164.68	HTTP	426	GET /static/faqs.css HTTP/1.1

2. Explanation:

Exercise 1:

Source: 192.168.191.52 (IP address of computer)

Destination: 128.119.245.12 (IP address of the website *gaia.cs.umass.edu*)

Exercise 2:

Source: 192.168.191.52 (IP address of computer)

Destination: 199.231.164.68 (IP address of the website *www.faqs.org*)

Question 9

1. Evidence:

Exercise 2:

No.	Time	Source	Step 1:	Destination	Protocol	Length	Info
409	3.892189	192.168.191.52		199.231.164.68	HTTP	528	GET /faqs/ HTTP/1.1
452	4.161206	199.231.164.68	Step 3:	192.168.191.52	HTTP	372	HTTP/1.1 200 OK (text/html)



Step 4:



Internet RFC Index		Search the FAQ Archives		Popular Usenet FAQs			
Usenet FAQ Index		ENHANCED BY Google Search		FAQs A-D FAQs J-P			
Other FAQs							
529	4.743157	199.231.164.68	Step 5:	192.168.191.52	HTTP	1302	HTTP/1.1 200 OK (PNG)
539	4.750089	199.231.164.68		192.168.191.52	HTTP	346	HTTP/1.1 200 OK (JPEG JFIF image)
541	4.770825	199.231.164.68		192.168.191.52	HTTP	717	HTTP/1.1 200 OK (PNG)

2. Explanation:

Step 1 – Send GET request:

- The computer sends a packet with the **HTTP GET method** to the server to request the web page content.

Step 2 – Server processing:

- The server receives the request and **processes** it.

Step 3 – Return HTTP Response:

- The server sends back an **HTTP Response packet**.
- If the page is loaded successfully, the **status code is 200**.



Step 4 – Display the web page:

- At this point, the user can **view the web page**.

Step 5 – Load additional objects:

- The browser continues to send **GET requests** for additional objects such as images, CSS, and JavaScript files.
- The server returns **HTTP Responses** for each object, and this process continues until all objects are loaded.

Expand

1. Evidence:

```
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  . : 
IPv6 Address. . . . .           : 2402:9d80:350:c24e:72d1:2221:c65c:676a
Temporary IPv6 Address. . . . . : 2402:9d80:350:c24e:943c:5941:d54a:acc4
Link-local IPv6 Address . . . . . : fe80::88fe:9b5:cf4d:6817%5
IPv4 Address. . . . .           : 192.168.191.52
Subnet Mask . . . . .           : 255.255.255.0
Default Gateway . . . . .       : fe80::7069:c2ff:fee4:ac84%5
                                   192.168.191.222
```

Exercise 1:

```
C:\Users\USER>ping gaia.cs.umass.edu

Pinging gaia.cs.umass.edu [128.119.245.12] with 32 bytes of data:
Reply from 128.119.245.12: bytes=32 time=360ms TTL=33
Reply from 128.119.245.12: bytes=32 time=391ms TTL=33
Reply from 128.119.245.12: bytes=32 time=480ms TTL=33
Reply from 128.119.245.12: bytes=32 time=397ms TTL=33

Ping statistics for 128.119.245.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 360ms, Maximum = 480ms, Average = 407ms
```

Exercise 2:

```
C:\Users\USER>ping www.faqs.org

Pinging faqs.org [199.231.164.68] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 199.231.164.68:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```



2. Explanation:

IP address use for:

- **Device Identification and Location:** Each device on a network has a unique IP address to distinguish it from others. Acts like a digital mailing address, ensuring data reaches the correct device.
- **Data Transmission and Routing:** Routers use IP addresses to route packets across networks to the correct destination. Supports two-way communication, ensuring data is sent and received accurately.
- **Network Configuration and Management:** IP addresses help configure firewalls, VPNs, and DHCP, enhancing network management and security. Allow administrators to monitor, control, and block unauthorized access.
- **Domain Name Resolution (DNS):** IP addresses work with DNS to map human-readable domain names (e.g., google.com) to numerical IPs, allowing devices to locate websites and services.

Check the IP address of computer:

1. Open Command Prompt (CMD).
2. Type the command: **"ipconfig"**
3. Press Enter.
4. Look for the line **"IPv4 Address"** – this shows the IP address of computer.

Check the IP address of a website

1. Open Command Prompt (CMD).
3. Type the command: **"ping <domain name>"**
4. Press Enter.
5. The response shows the IP address of the website in **square brackets**

END